Linear Discriminant Analysis

Machine Learning Techniques

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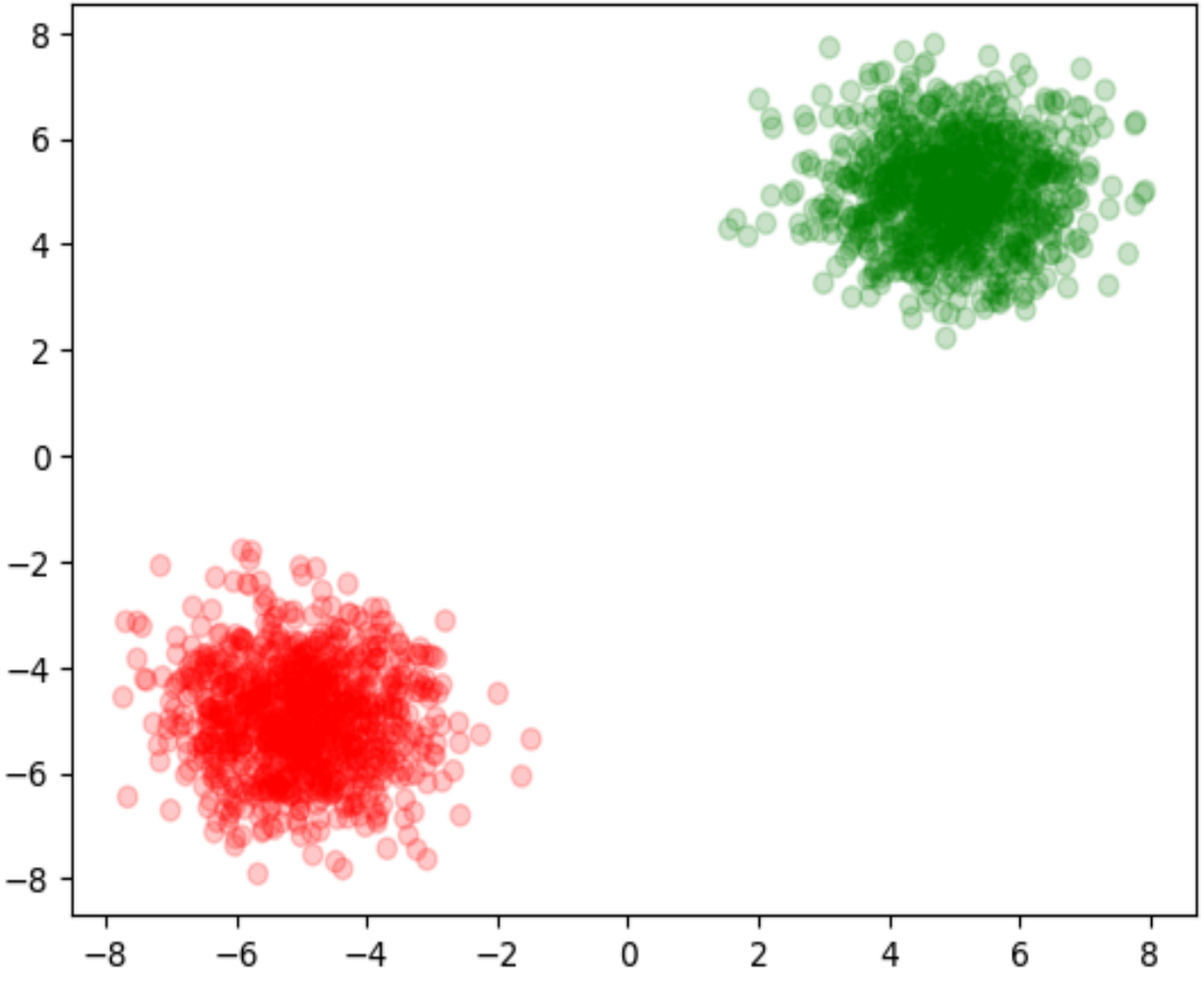
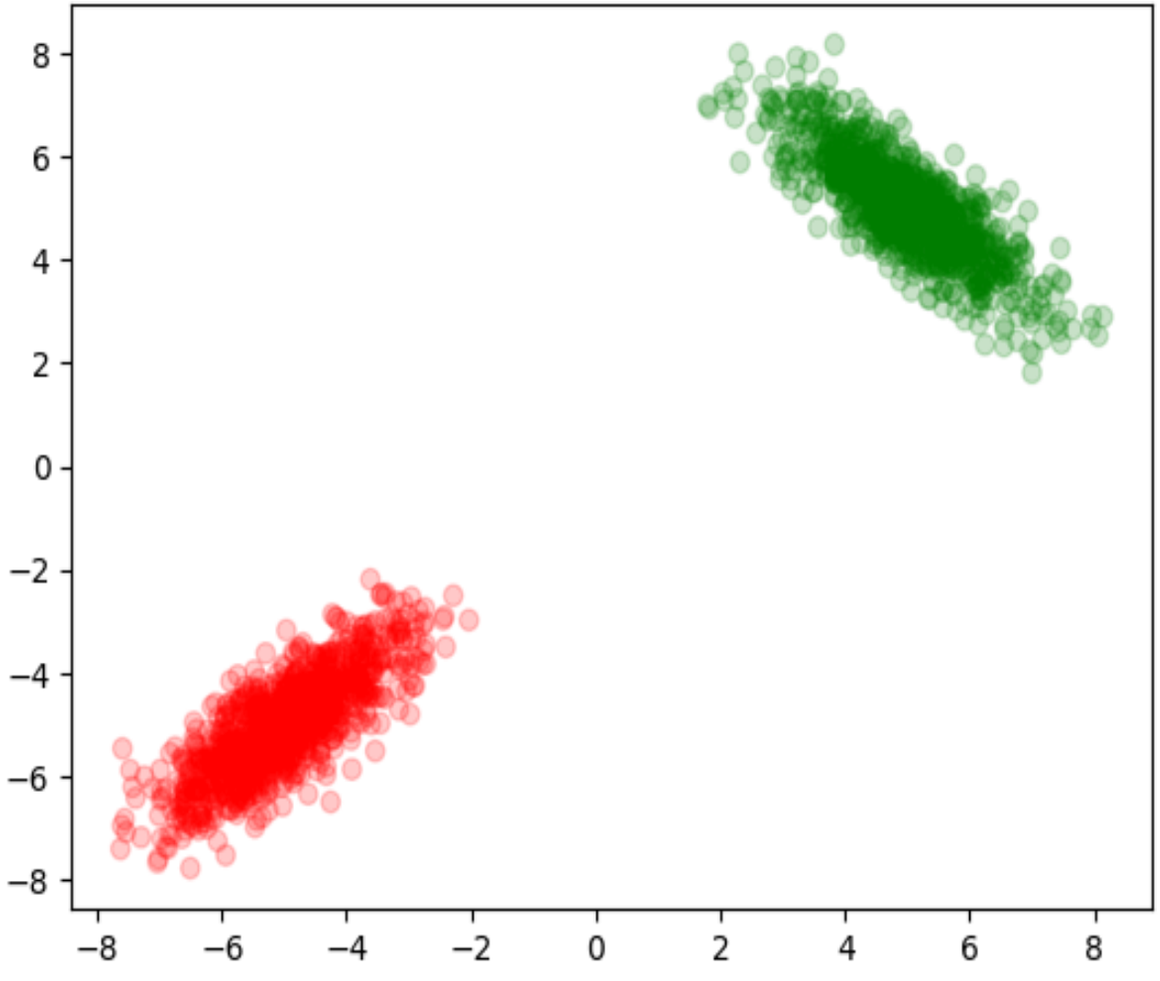
Linear/Quadratic Discriminant Analysis

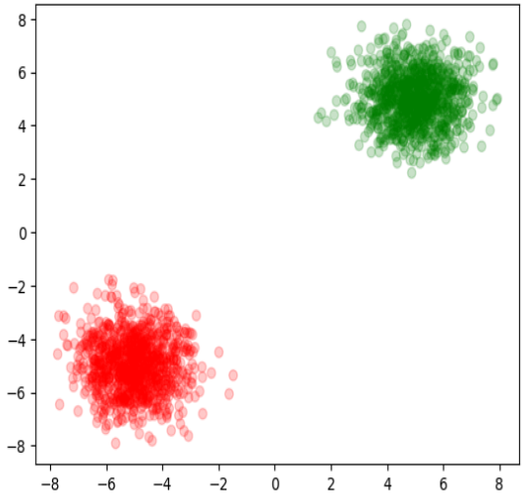


Identical Covariance Matrices



Different Covariance Matrices

Linear Discriminant Analysis

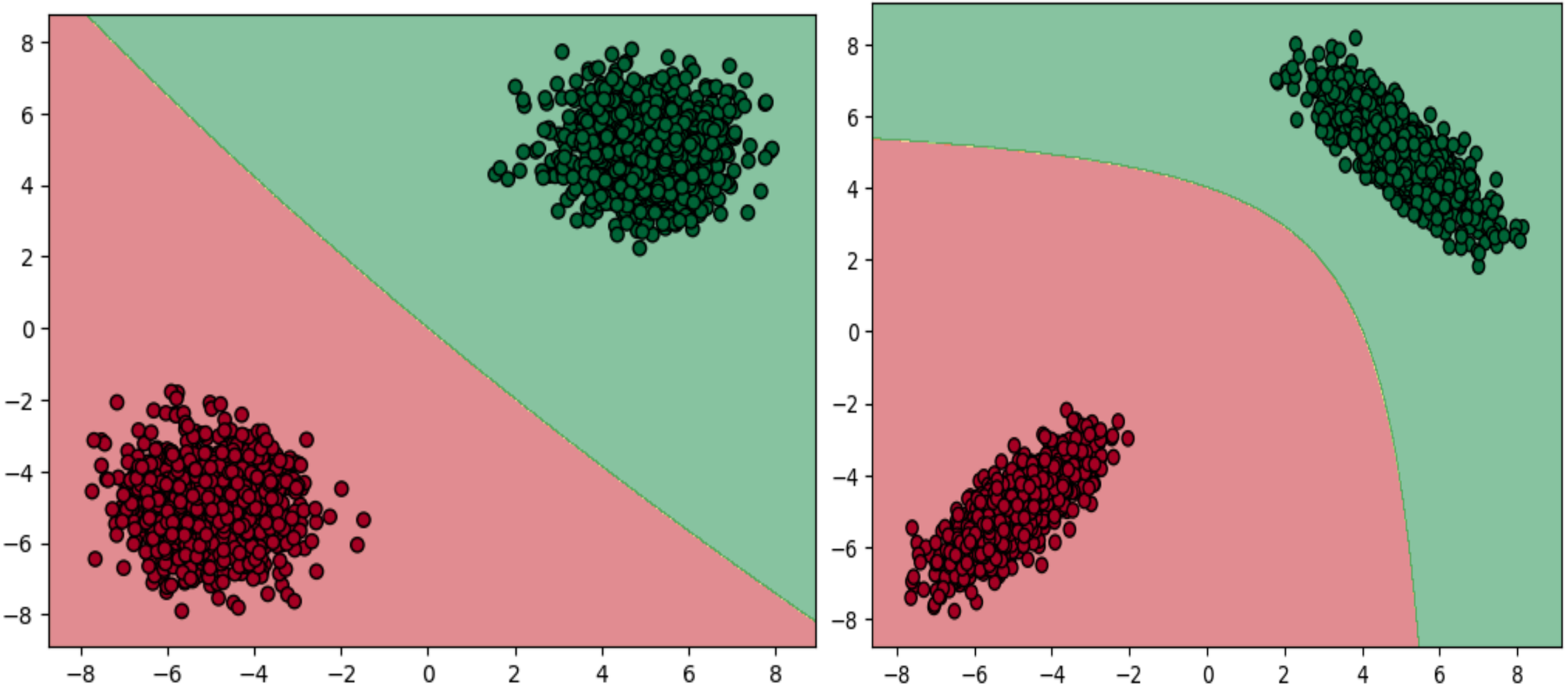


$$\mathbf{x} \mid y = 0 \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma})$$

$$\mathbf{x} \mid y = 1 \sim \mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma})$$

$$\boldsymbol{\mu}_0 = \frac{\sum_{i=1}^{n} \mathbb{I}(y_i = 0)\mathbf{x}_i}{\sum_{i=1}^{n} \mathbb{I}(y_i = 0)}$$

$$\boldsymbol{\Sigma} = \frac{1}{n}\sum_{i=1}^{n} (\mathbf{x}_i - \boldsymbol{\mu}_{y_i})(\mathbf{x}_i - \boldsymbol{\mu}_{y_i})^T$$

$$\boldsymbol{\mu}_1 = \frac{\sum_{i=1}^{n} \mathbb{I}(y_i = 1)\mathbf{x}_i}{\sum_{i=1}^{n} \mathbb{I}(y_i = 1)}$$

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